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Where Minds Pollinate

Class 12
Pakistan Studies

Chapter 4
Solved Exercise

National Book Foundation

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Though Pakistan has policies for climate change mitigation, economic and political will prevent effective measures from being implemented fully, hampering efforts to control climate change.

12. Poverty and Socioeconomic Factors

Poverty restricts people's ability to adopt environmentally friendly practices. Many poor communities depend on wood fuel for cooking and heating, which increases deforestation and pollution.

Conclusion

In conclusion, climate change in Pakistan is driven by a combination of environmental, economic and social factors. Deforestation, reliance on fossil fuels, industrial emissions, and rapid urbanization are major contributors. Melting glaciers, water management issues, and natural disasters further worsen the situation. Despite challenges, raising awareness, improving policies, and adopting clean technologies are essential steps to mitigate climate change. Pakistan must work collectively—government, communities, and individuals—to protect its environment for future generations and reduce the adverse effects of climate change.

Section 2 Geography of Pakistan

Unit 04: Natural Vegetation and Forests of Pakistan

SHORT QUESTIONS

Q.1 Identify different types of forests.

Forest Type	Area (Region in Pakistan)	Height Above Sea Level
Alpine Forests	Gilgit-Baltistan, Northern Kashmir	Above 3,500 meters

Forest Type	Area (Region in Pakistan)	Height Above Sea Level
Sub-Alpine Forests	Upper Swat, Chitral, Astore	3,000 – 3,500 meters
Coniferous Forests	Murree, Swat, Dir, Kaghan Valley	1,000 – 3,000 meters
Sub-Tropical Pine Forests	Lower hills of KP, Punjab, Balochistan	900 – 1,700 meters
Scrub Forests	Potohar Plateau, Salt Range	600 – 1,200 meters
Riverine Forests	Along Indus River (Sindh & Southern Punjab)	0 – 200 meters
Tropical Thorn Forests	Southern Punjab, Sindh, Balochistan plains	Below 500 meters
Mangrove Forests	Indus Delta, Karachi coast	Sea level (0 – 10 meters)
Irrigated Plantations	Central Punjab (Changa Manga, Chichawatni)	100 – 300 meters

Q.2 Why irrigated forests are important for Pakistan?

1. Irrigated forests provide a sustainable source of wood for construction, furniture, and fuel.
This reduces pressure on natural forests and supports local economies.
2. They help reduce soil erosion, improve groundwater recharge, and purify air.
These forests also act as windbreaks and help in stabilizing the climate.
3. Forest management, harvesting, and related industries create jobs for rural communities.
This helps in poverty reduction and supports local development.
4. Irrigated forests serve as habitats for birds and small animals.
They help conserve biodiversity in areas with otherwise limited natural vegetation.

Q.3 Compare Productive and Protective forests.

1. **Main Purpose**
Productive forests are managed mainly to produce timber, fuelwood, and other forest products.
Protective forests are preserved to protect the environment, such as preventing soil erosion and floods.
2. Productive forests contribute directly to the economy through wood and non-wood forest products.
Protective forests provide indirect benefits like clean air, water conservation, and climate control.
3. Productive forests are often found in accessible areas with good growth conditions for trees.
Protective forests are usually located in sensitive areas like hills, watersheds, or coastal zones.
4. Productive forests are more intensively managed and regularly harvested.
Protective forests have restricted human activities to maintain their natural functions.

Q.4 Identify the government efforts to protect the environment.

1. **Billion Tree Tsunami Project** – A massive tree plantation drive to restore forests and combat climate change.
2. **Clean Green Pakistan Initiative** – A campaign to promote cleanliness, tree plantation, and proper waste management.
3. **Environmental Protection Laws** – Laws like the Pakistan Environmental Protection Act (1997) to control pollution.
4. **National Climate Change Policy** – A strategy to deal with climate challenges through conservation and green energy.

Q.5 Assess the importance of Mangroves for Pakistan.

1. Mangroves protect the coastline from erosion, storms, and floods.
2. They provide habitat and breeding grounds for fish and marine life.
3. Mangroves help reduce climate change by storing carbon.
4. They support local fishing communities and preserve biodiversity.

Q.6 What are the effects of deforestation?

1. Causes loss of wildlife habitat and decreases biodiversity.
2. Leads to soil erosion and loss of fertile land.
3. Increases greenhouse gases, contributing to climate change.
4. Disrupts water cycles, causing floods and droughts.

Q.7 Why are Mangroves important for fishing?

1. Mangroves provide safe breeding and nursery areas for many fish species.
2. They offer shelter and food for young fish, increasing their survival rates.
3. Mangroves help maintain healthy marine ecosystems that support fisheries.
4. They protect coastal waters from pollution and erosion, benefiting fishing habitats.

Additional Short Questions

Q1. Give Pakistan' exact area in square km and its location on the globe with reference to latitude and longitude?

1. Pakistan covers an area of 796096 km square.
2. It is situated at 61"-75.5" East longitude and 24.5"-36.5" North Latitude.
3. It occupies 18.78% of total area of South Asia.
4. The land stretch of Pakistan from East to West is 850 km and from North to South is 1600 Km.

Q2. Describe Pakistan's position with reference to her neighbouring countries? (Give the length of the common boundaries shared with neighbours).

Country	Length of border	Direction w.r.t Pakistan
Afghanistan	2612 km	North west
India	2110 km	East
Iran	909 km	West & South West
China	600 km	North & North East
Arabian Sea	1059 km	South
Wakhan	20 km	North West

Q3. What are three main classifications of the land of Pakistan with reference to physical features?

1. Mountains:

There are three categories of mountains in Pakistan;

- i) Himalayas range ii) Karakoram range iii) Hindukush range

2. Plateaus:

There are two plateaus in Pakistan;

- I. Pothohar plateau II. Baluchistan plateau

3. Plains:

There is one plain in Pakistan; Indus Plain which is further divided into two parts; i) Upper Indus plain ii) Lower Indus plain

Q4. Write the names of the mountains (as well as mountainous ranges) lying to the North West and West of Pakistan.

North Western Mountains Hindukush Range
Waziristan Hills
Koh Sufaid
Swat and Chitral Hills

Western Mountains Suleman Hills
Kirthar Mountains

Q5. Enlist the mountains forming the Himalayas Range.

Himalayas are divided into three categories;

Greater Himalayas	(ranges above 6000 m of height)
Lesser Himalayas	(ranges between 4000 m to 6000 m of height)
Shiwalik Range	(ranges below 4000m of height)

Greater Himalayas are also called Central Himalayas.

Q6. Write a note on the climates of Pakistan.

Pakistan is situated in Monsoon region as the land of Pakistan is divided into four different categories similarly its climate varies with the change in altitude and position;

Sub-tropical highland region	Extremely cold
Sub-tropical lowland region	More or less warm and cold
Baluchistan Type of plateau	Rocky terrain, extremely hot at day time and cold at night
Coastal areas	Moderate temperature, neither very cold nor very hot

Q7. Write a note on the weather of the mountainous regions of continental climates.

1. North and North Western Mountainous region receive rainfall during summer (93 mm per year). Western Mountainous region receives more rainfall during summer (250 mm per year).
 2. Southern Mountainous region receives more rainfall in summer (125 mm per year).
 3. Shiwalik Mountains receives more rainfall in summer and less in winter (155 mm per year).
- Average rainfall in Pakistan is 175 mm per year.

Q8. Indicate the difference of population between different regions of Pakistan.

1. Population of Pakistan is divided into four provinces

Punjab (26% of total area of Pakis)	7.6 crore (66% of total population of Pakistan)
Sindh	3.41 crore
KPK	1.77 crore
Baluchistan	65 lac (5% of total population of Pakistan)

These statistics show the uneven distribution of population in Pakistan.

Q9. Write a note on the imbalance of the economic growth between different regions of Pakistan.

Allah Almighty has blessed Pakistan with innumerable natural resources. It is observed that people settle at places where there is more economic activity, business and job opportunity. Sind has the highest per capita income which is one and a half times higher than Punjab, three times higher than KPK and four times higher than Baluchistan. So, there are lots of imbalances of economic growth in Pakistan.

Q10. Bring about the central position Pakistan enjoys as a significant state of the Muslim world.

1. Pakistan is situated in the middle of the Muslim world. To the North and North West lies Afghanistan and to the North Central Asian States, to the South West lies Iran and other Middle Eastern countries and to the East lies Bangladesh and Eastern Countries (Indonesia and Malaysia).
2. Pakistan is called "The Heart of the Muslim World". Being an active member of ECO and atomic power, Pakistan is a significant state of the Muslim World.

Q11. Enlist three main objectives of tourism.

The main objectives of tourism are:

1. Research and fact findings.
2. Attainment of knowledge and information.
3. Recreation.

Pakistan is bestowed with enormous natural tourists cites like Kaghan, Naran, Kalam, Sawat and Saiful malook in Northern areas and Ziarat in Baluchistan etc.

There are so many others in different regions of Pakistan for tourists' attraction.

Q12. Under what major heads, important sites of tourist attraction can be studied.

Tourist attraction in Pakistan can be studied under the following headings:

1. Archaeological treasure.
2. Historical buildings.
3. Health and recreation resorts.

Moenjodaro, Harappa and excavations at different places in Pakistan are precious archiological treasures. Badshahi mosque, Shahi Qala, Jahangir's Tomb, Rohtas Fort and Balahasar Fort are some important historical buildings. Nathia Gali, Ayyubia, Murree, Patriata, Ziarat are some important health and recreation resorts.

Q13. Enlist important hill resorts of the Punjab and the KPK.

The important hill resorts of the Punjab and the KPK are:

1. Patriata which is about 25 km away from Murree. (Punjab)
2. Murree which is about 85 km from Rawalpindi. (Punjab)
3. Pir Sohawa in Margalla Hills.
- 4 Nathiya Gali which is 33 km away from Murree. (KPK) Ayyubia is 27 km from Murree. (KPK)

LONG QUESTIONS

Q.1 Explain with reference to examples, how deforestation has affected the land and the economy of the country?

1. Loss of Wildlife Habitat

Deforestation destroys the natural homes of many species, leading to reduced biodiversity. For example, the depletion of forests in northern Pakistan has threatened many native animals.

2. Soil Erosion

Without trees, soil is easily washed away by rain, causing erosion. In hilly areas like Murree, deforestation has increased landslides and soil loss.

3. Decrease in Agricultural Productivity

Eroded and degraded soil reduces crop yields and farm income. Farmers in deforested regions of Khyber Pakhtunkhwa face lower productivity due to poor soil.

4. Water Cycle Disruption

Trees help maintain the water cycle by absorbing and releasing water. Deforestation in parts of Punjab has led to reduced rainfall and water shortages.

5. Flooding

Loss of forest cover increases surface runoff, causing floods.

Floods in Sindh and Punjab have worsened after upstream deforestation.

6. Climate Change Contribution

Cutting trees releases stored carbon dioxide, increasing greenhouse gases.

Pakistan's deforestation adds to global warming, affecting agriculture and health.

7. Decline in Timber and Forest Products

Reduced forest area lowers the supply of wood, fuel, and medicinal plants.

This has increased dependence on imports and raised costs for local industries.

8. Impact on Livelihoods

Many communities rely on forests for income and resources.

Deforestation has caused job losses and poverty in rural forest-dependent areas.

9. Loss of Tourism Potential

Forests attract tourists for hiking and nature activities.

Areas like Kaghan Valley face reduced tourism due to deforestation.

10. Increase in Natural Disasters

Deforested regions are more prone to landslides, droughts, and storms.

The 2005 Kashmir earthquake impacts worsened partly due to forest loss.

11. Reduced Carbon Sequestration

Fewer trees mean less carbon dioxide is absorbed from the atmosphere.

This weakens efforts to combat climate change nationally.

12. Economic Burden on Government

Conclusion

Deforestation has severely affected Pakistan's land through soil erosion, water scarcity, and natural disasters, which in turn harm agriculture and livelihoods. Economically, it increases costs, reduces forest product supply, and damages tourism. Sustainable forest management is essential to protect the environment and support the country's economy.

Q.2 Analyze the importance of afforestation for the economic and social needs of the people.

1. Employment Generation

Afforestation projects create jobs in planting, maintenance, and forest management.

This helps reduce unemployment, especially in rural areas.

2. Source of Timber and Fuelwood

New forests provide sustainable timber and fuelwood supplies.

This supports construction industries and meets household energy needs.

3. Improvement in Agricultural Productivity

Trees improve soil fertility and reduce erosion, boosting crop yields.

Farmers benefit from better soil and increased food production.

4. Climate Regulation

Afforestation helps absorb carbon dioxide, reducing global warming impacts.

Stable climate conditions benefit agriculture and daily life.

5. Water Conservation

Trees help retain groundwater and maintain river flows.

This ensures water availability for drinking and irrigation.

6. Control of Soil Erosion

Tree roots hold soil in place, preventing landslides and loss of fertile land.

This protects farmland and infrastructure.

7. Reduction in Natural Disasters

Forests act as buffers against floods, storms, and droughts.

They safeguard lives and property, reducing disaster-related costs.

8. Improved Air Quality

Trees filter pollutants and produce oxygen.

This leads to healthier communities and fewer respiratory diseases.

9. Support for Biodiversity

Afforestation restores habitats for wildlife and plants.

This preserves ecosystems that are vital for environmental balance.

10. Promotion of Eco-Tourism

Forested areas attract tourists, creating income for local people.

Eco-tourism supports community development and cultural exchange.

11. Social Well-being and Recreation

Green spaces improve mental health and provide recreational areas.

Communities benefit from better quality of life and social cohesion.

12. Sustainable Economic Growth

Afforestation supports industries like paper, furniture, and herbal products.

It promotes long-term economic stability by preserving natural resources.

Conclusion:

Afforestation is vital for meeting both economic and social needs by creating jobs, protecting the environment, and improving quality of life. It supports sustainable development, ensuring resources and well-being for present and future generations.

Q.3 Evaluate the role of Mangrove forests in mitigating climate change.

1. Carbon Storage

Mangroves store large amounts of carbon in their roots, trunks, and soil.

This helps reduce carbon dioxide in the atmosphere.

2. Long-Term Carbon Sequestration

The carbon stored in mangrove soils can remain trapped for hundreds of years.

This makes mangroves very effective in fighting climate change.

3. Reducing Greenhouse Gases

By absorbing carbon dioxide, mangroves lower greenhouse gas levels.

This helps slow down global warming.

4. Protecting Coastal Areas

Mangroves act as natural barriers against storms and rising sea levels.

This protects communities from climate-related disasters.

5. Preventing Soil Erosion

Mangrove roots hold soil and prevent erosion caused by waves and floods.

This keeps coastal land stable and healthy.

6. Supporting Biodiversity

Healthy mangroves provide homes for many plants and animals.

Biodiversity helps ecosystems stay strong and adapt to changes.

7. Improving Water Quality

Mangroves filter pollutants and trap sediments from rivers.

Clean water supports marine life and reduces climate stress.

8. Reducing Ocean Acidification

Mangrove ecosystems help balance the acidity of coastal waters.

This protects coral reefs and marine species affected by climate change.

9. Supporting Fisheries

Mangroves provide nursery grounds for fish and seafood.

This supports food security for coastal communities.

10. Helping Local Livelihoods

Mangrove forests provide resources like wood and honey sustainably.

This reduces pressure on other forests and supports communities.

11. Promoting Eco-Tourism

Mangroves attract tourists interested in nature and wildlife.

Tourism creates awareness and economic benefits for conservation.

12. Encouraging Climate Policies

Mangrove conservation is included in many climate action plans worldwide.

Protecting mangroves supports global efforts to combat climate change.

Conclusion

Mangrove forests play a key role in reducing climate change by storing carbon, protecting coastlines, and supporting ecosystems. Their conservation is essential for a healthy environment and the well-being of people living near the coasts.

Q.4 Identify the need of different geographical tools and techniques to map different types of forests.

1. Aerial Photography

Used to capture detailed images of forests from above.

Helps identify forest boundaries and types.

2. Satellite Imagery

Provides large-scale, up-to-date pictures of forest areas.

Useful for monitoring changes like deforestation or growth.

3. GPS (Global Positioning System)

Helps in precise location mapping of forest regions.

Essential for tracking specific forest sites accurately.

4. Remote Sensing

Uses satellite or aerial data to study forests without physical presence.

Allows analysis of forest health, density, and changes over time.

5. GIS (Geographic Information Systems)

Integrates data from various sources to create detailed forest maps.

Useful for managing forest resources and planning conservation.

6. Topographic Maps

Show elevation and terrain features of forested areas.

Helps understand forest types related to altitude and slope.

7. Soil Sampling Techniques

Analyze soil quality in different forest zones.

Important for classifying forests based on soil type.

8. Vegetation Surveys

On-ground studies to identify tree species and forest density.

Provides detailed data about forest composition.

9. Climate Data Collection

Records temperature, rainfall, and humidity in forest areas.

Helps classify forests based on climatic conditions.

10. Drones

Provide high-resolution images and videos of forests.

Useful for mapping inaccessible or dense forest areas.

11. Forest Inventory Methods

Count and measure trees to estimate forest resources.

Vital for sustainable forest management.

12. Hydrological Tools

Study water flow and sources in forest ecosystems.

Important for mapping riverine and mangrove forests.

Conclusion

Different geographical tools and techniques are essential to accurately map and monitor various forest types. They help in effective forest management, conservation, and understanding environmental changes.

Q.5 Evaluate the impact of natural vegetation and forests on the lifestyles of people.

1. Source of Food

Forests provide fruits, nuts, and edible plants that support local diets. Many communities rely on forest products for daily nutrition.

2. Fuel and Energy

Wood from forests is used as firewood and charcoal for cooking and heating. Especially important in rural areas without electricity.

3. Medicinal Plants

Natural vegetation offers herbs and plants used in traditional medicine. These help treat illnesses and promote health in many communities.

4. Livelihood Opportunities

Forests create jobs in logging, collection of non-wood products, and eco-tourism. Many people earn income through forest-related activities.

5. Cultural Significance

Forests hold spiritual and cultural value for many indigenous groups. They are places for rituals, festivals, and community gatherings.

6. Building Materials

Wood and bamboo from forests are used in constructing homes and tools. Local architecture often depends on available natural materials.

7. Climate Regulation

Forests help regulate local weather, making life comfortable. They provide shade, reduce heat, and improve air quality.

8. Water Resources

Vegetation maintains water cycles and ensures clean water supply. Communities depend on forests for rivers and groundwater recharge.

9. Protection from Natural Disasters

Forests reduce risks from floods, landslides, and storms. This safeguards villages and farmland.

10. Recreation and Tourism

Forests offer places for hiking, picnics, and nature-based tourism. This promotes leisure and boosts local economies.

11. Soil Fertility

Leaf litter and organic matter improve soil quality for farming. Farmers benefit from healthier soils near forests.

12. Air Quality and Health

Trees clean the air by absorbing pollutants and releasing oxygen. Better air means fewer respiratory problems for people.

Conclusion

Natural vegetation and forests deeply influence people's lifestyles by providing resources, protecting the environment, and supporting cultural values. Their preservation is vital for sustainable living and community well-being.